COURSE OUTCOMES

- Demonstrate a solid understanding of the basic types of recommender systems and their real-world applications.
- Identify the strengths and limitations of various recommendation techniques.
- Stay up to date with current research in recommender systems.
- Gain the foundational skills and practical experience required to work on recommender systems in industry or research, positioning students for roles in AI, machine learning, or data science.

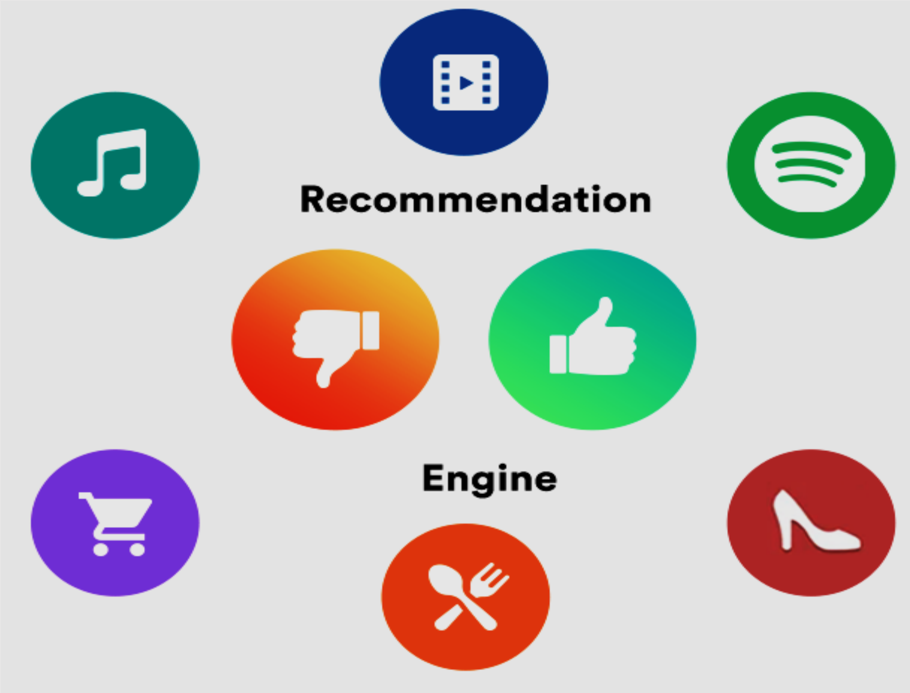
Part 1 Introduction to RecSys

Part1: Introduction to Recommender systems

1. What are Recommender Systems (RecSys)?
2. Where to Apply/Find Recommender Systems?
3. Types of Recommender Systems
4. Advantages of Recommender Systems
5. Challenges in Recommender Systems
6. Future Directions in Recommender Systems

Recommender Systems (RecSys)

- A **recommender system / recommendation system** is a type of software or algorithm designed to predict and suggest items that a user might find interesting.



Recommender Systems (RecSys)

- A **recommender system** is a subclass of information filtering systems that predict and suggest relevant items to users based on their preferences, past behavior, and other data points.
- These systems are widely used in various industries to provide personalized recommendations, improving user experience and boosting engagement by suggesting products, services, or content based on the user's preferences, behavior, and interactions.

Where to Apply/Find Recommender Systems

- Recommender systems are found in many industries, including:

E-commerce

Amazon, eBay, and Alibaba use recommendation algorithms to suggest products to users based on their previous purchases or browsing history.

Streaming Services

Netflix, YouTube, Spotify, and Hulu recommend movies, TV shows, videos, and music based on a user's preferences, watch/listen history, and similarity to other users.

Education

Systems like Coursera, Udemy, and edX use recommendation systems to suggest courses based on the user's learning history or goals.

Social Media

Facebook, Instagram, Twitter, and TikTok use recommendation systems to personalize content, suggest friends.

News/Advertising

Platforms like Google News and Flipboard recommend news articles based on user interests and reading patterns, and display relevant ads.

Types of Recommender Systems

- Recommender systems can be broadly classified into several types based on the techniques they use :

Collaborative Filtering

The system recommends items based on user interactions with other items. Two primary forms exist

User-based collaborative filtering: Recommends items based on similarities between users.

Item-based collaborative filtering: Recommends items based on similarities between items.

Example: Netflix recommending a movie because similar users watched it.

Content-Based Filtering

This method uses the characteristics of the items and compares them to the user's preferences. The system recommends items similar to what the user has liked before.

Example: Spotify recommending songs that are similar to the ones you have listened to.

Hybrid Models

These combine multiple recommendation approaches, such as combining content-based and collaborative filtering techniques, to improve accuracy.

Example: Netflix uses a hybrid system that takes into account both content-based and collaborative methods to suggest content.

Types of Recommender Systems

❑Modern Era

Knowledge-based Systems

These systems make recommendations based on the user's needs and knowledge of how items meet those needs.

Example: A travel website suggesting vacation packages based on user preferences and budget constraints.

Deep Learning-based Models

Modern recommender systems use neural networks, convolutional neural networks (CNNs), or recurrent neural networks (RNNs) to learn complex patterns in user behavior, preferences, and the attributes of items.

Example: YouTube recommending videos by analyzing user interaction, viewing patterns, and metadata.

Context-aware Systems

These systems make recommendations by considering contextual information such as location, time, or device type.

Example: Suggesting a nearby restaurant while a user is traveling.

Advantages of Recommender Systems

Recommender systems provide personalized content or product suggestions, improving user satisfaction and engagement. **Personalization**	Helps users discover new items, saving them time and effort. **Improved User Experience**
By offering relevant recommendations, businesses can increase sales or user engagement. **Increased Sales/Engagement**	Efficiently sifts through large catalogs of items to offer personalized suggestions, particularly useful for platforms with vast amounts of content (e.g., Netflix, Amazon). **Handling Large Catalogs**

Challenges of Recommender Systems

Cold Start Problem

Recommender systems struggle to make accurate predictions for new users or new items with little to no historical data.

Bias

Recommender systems can reinforce existing biases, recommending only popular or trending items and ignoring niche content.

Privacy Concerns

Collecting user data to make recommendations can raise privacy issues, especially if users are unaware of the data being used.

Data Sparsity

Many recommendation algorithms, especially collaborative filtering, require sufficient user-item interactions. In many cases, there is insufficient data to make reliable recommendations.

Overfitting

Algorithms may become too specific, overfitting user preferences and missing opportunities to recommend diverse or chance in a happy content.

Scalability

As the number of users and items grows, it becomes challenging to maintain real-time recommendations without significant computational resources.

Future Directions in Recommender Systems

Deep Learning Approaches

The use of neural networks and deep learning allows recommender systems to capture complex patterns in data, improving personalization and accuracy.

Context-aware Recommendations

Incorporating more contextual information, such as time, location, and user activity, will make recommendations more dynamic and relevant.

Explainable Recommendations

As users demand more transparency, there is a growing focus on providing explainable recommendations that show why an item is suggested.

Cross-Domain Recommendations

Recommender systems may evolve to provide recommendations across different platforms or services, improving personalization across various domains (e.g., integrating recommendations between an e-commerce and a social media platform).